

Ryan Cho

Full Stack Software Engineer (Java & React)

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PROFESSIONAL SUMMARY

Software engineer with 6+ years building full stack applications in financial services and capital markets. Skilled in Java/Spring Boot microservices, React/TypeScript frontends, and REST APIs for payment workflows, order management, and marketplace fulfillment. Delivers scalable cloud-native services with PostgreSQL/MySQL, event-driven patterns, and CI/CD. Strong backend architecture and responsive UI development in Agile, cross-functional teams.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Owned **Java Spring Boot microservices** for payment initiation, settlement reconciliation, and account funding across a multi-tenant capital markets platform serving institutional clients.
- Built **React and TypeScript** operator dashboards for order tracking, transaction history, and fulfillment status, giving teams real-time visibility into end-to-end payment lifecycles.
- Designed **REST API contracts and webhook handlers** integrating third-party payment processors, standardizing callback processing and reducing manual exception handling for operations teams.
- Optimized **PostgreSQL** schemas and query paths for high-volume transaction ledgers, maintaining reliable read performance as monthly record volume scaled across product lines.
- Introduced **event-driven workflows** on AWS SQS, SNS, and Lambda to propagate async payment notifications, decoupling fulfillment services from core order processing.
- Established **Docker-based CI/CD pipelines** with automated integration tests, enabling frequent releases and zero-downtime deployments across staging and production environments.
- Partnered with compliance stakeholders to implement **PCI-aware data handling** patterns for card and ACH flows, keeping sensitive fields out of application logs and audit trails.
- Mentored engineers on **microservices design**, API versioning, and React component architecture, improving code review throughput and onboarding for new team members.
- Hardened **order and cart orchestration** logic for concurrent checkout scenarios, preventing duplicate charges and improving reliability of marketplace-style purchase flows.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **Java Spring Boot services** supporting trade settlement, cash movement, and internal payment routing for global banking operations.
- Delivered **React interfaces** for operations teams to monitor transaction queues, exception cases, and fulfillment status across cross-border payment channels.
- Designed **RESTful integration layers** connecting core banking systems with vendor payment APIs, normalizing request and response models for downstream microservices.
- Maintained **MySQL and PostgreSQL** data models for transaction audit trails, preserving referential integrity across order and payment state transitions.
- Built webhook and message listeners on **AWS API Gateway and Lambda** to broadcast payment status events to consuming teams with near-real-time delivery.
- Contributed to **microservices decomposition** of a legacy monolith, extracting payment validation and order matching into independently deployable services.
- Participated in **Agile sprint delivery** and Git trunk-based development, coordinating releases with QA and production support across distributed teams.

- Applied **JVM profiling and batch tuning** to shorten settlement job runtimes during peak trading volumes, improving overnight processing reliability.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **React and JavaScript** components for customer order history and payment receipt views within a B2B marketplace prototype.
- Implemented **REST API** consumption with structured error handling and loading states for checkout and cart management screens.
- Authored unit tests and participated in **Git pull request** reviews, aligning TypeScript modules with team coding standards.
- Assisted backend engineers with **PostgreSQL** validation scripts to verify order and inventory data during feature testing cycles.
- Supported integration testing of **payment callback endpoints**, documenting webhook payload formats for QA and release handoff.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Developed responsive **HTML, CSS, and JavaScript** pages for nonprofit fundraising campaigns using mobile-first layout patterns.
- Integrated third-party **donation payment widgets** with client-side validation to support secure online contribution flows.
- Collaborated with stakeholders in **Agile check-ins** to iterate on site structure, content priorities, and release timelines.
- Published static assets and configuration updates through a **Git-based release workflow** to production hosting environments.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React.js, TypeScript, JavaScript, HTML5, CSS3, Responsive UI, Redux, Component Architecture

Backend: Java, Spring Boot, REST APIs, Microservices, Event-Driven Architecture, Node.js

Databases: PostgreSQL, MySQL, SQL, Schema Design, Query Optimization, Transaction Modeling

Cloud & DevOps: AWS (Lambda, SQS, SNS, API Gateway), Docker, Kubernetes, CI/CD, Git, Containerization

Monitoring & Tools: Application Performance Monitoring, Log Aggregation, JUnit, Postman, Swagger/OpenAPI

Workflow & PM: Agile/Scrum, Jira, Cross-Functional Collaboration, Technical Documentation, Code Review